MILITARY SPECIFICATION

ADAPTER UNIT, INPUT, NTDS INTERFACE

1. SCOPE

1.1 This specification covers the fabrication, performance, and test requirements for an Interface Adapter Unit to interface the AN/SRA-57, 58 transmitter multicoupler groups with either the AN/SRC-23 or AN/SRC-16 Naval Tactical Data Systems (NTDS) system. Each adapter shall interface four channels of an AN/SRA-57 or 58 coupler group.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400 - Electronic Equipment Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.4).

3.2 The NTDS Interface Adapter Unit shall be in accordance with MIL-E-16400 to the extent specified herein.

3.3 Items to be supplied. The following items shall be supplied by the contractor.

3.3.1 NTDS interface adapter unit. The NTDS Interface Adapter Unit shall consist of a card retainer box containing four plug-in printed circuit cards

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along with the necessary connectors and terminal boards to connect with both the NTDS and the AN/SRA-57, 58 transmitter multicoupler groups. Each plug-in printed circuit card shall contain all the components required for control of a single channel.

3.3.2 Installation kit. An installation kit consisting of necessary installation hardware, along with detailed installation and connection details shall be provided with each Interface Adapter Unit. Installation instructions shall detail AN/SRA-56, 57, 58 modifications.

3.3.3 Manuscript change to AN/SRA-56, 57, 58 technical manual. Two copies of manuscript change to AN/SRA-57, 58 Technical Manual shall be supplied with each Interface Adapter Unit.

3.3.4 A self-adhesive NOTICE identifying coupler group as modified and containing the NTDS Interface Adapter Unit to be affixed to the AN/SRA-57, 58 coupler group shall be provided with each Interface Adapter.

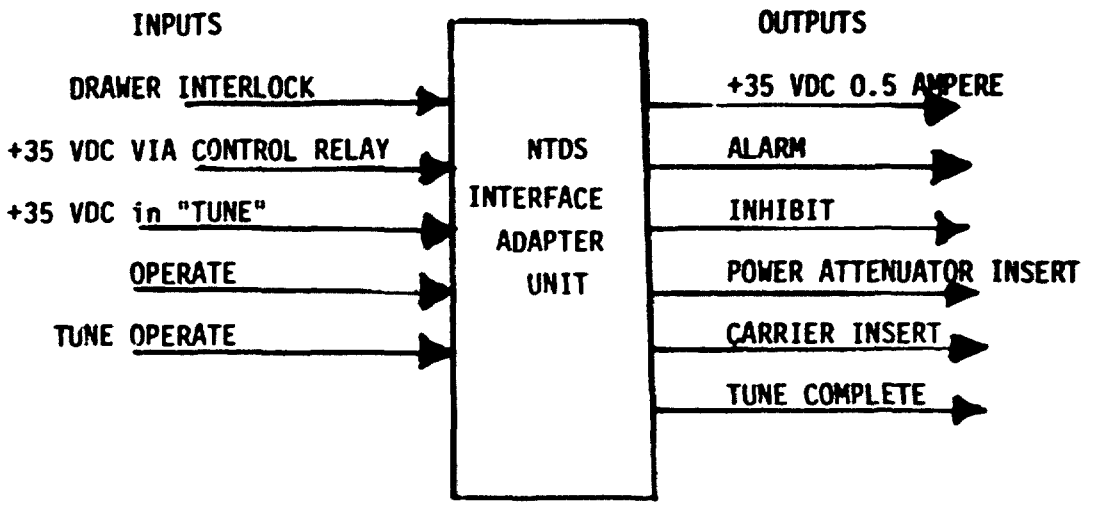
3.3.5 An instruction plate giving condensed operating instructions for the AN/SRA-57, 58 coupler groups fitted with the Interface Adapter shall be provided with each Interface Adapter.

3.3.6 The contractor shall assure compliance with identification and marking requirements in conformance with applicable requirements of MIL-E-16400. Nomenclature shall be requested.

3.4 Identification of inputs and outputs. For purposes of defining performance requirements, NTDS Interface Adapter inputs and outputs are identified:

INPUTS	OUTPUTS
Drawer interlock	+35 V dc 0.5 ampere
+35 V dc via control relay	ALARM
+35 V dc in "TUNE"	INHIBIT
OPERATE	Power Attenuator Insert
TUNE ACTIVATE	CARRIER INSERT
	TUNE COMPLETE

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NTDS INTERFACE UNIT BLOCK DIAGRAM

The above inputs and outputs are given for one of the four identical channels. The +35 V dc (0.5 ampere) is used to provide a source for the operation of the associated MX-4845A/SRA-34 coupler adapter in the event of loss of 400 Hz primary power.

3.4.1 Performance of each channel.

3.4.1.1 Apply a ground to the Drawer Interlock terminal, all other terminals open. Alarm terminal shall exhibit a resistance of less than 1 ohm to ground.

3.4.1.2 Apply ground to Drawer Interlock and the Operate terminals. Connect the +35 V dc via control relay terminal and the +35 V dc in tune terminal together and the pair to a source of +35 V dc. Alarm and Carrier Insert terminals shall show a resistance of less than 1 ohm to ground. Power Attenuator Insert terminal shall show +35 V dc Inhibit and Tune Complete terminals shall show an open circuit (in excess of 100,000 ohms to ground).

3.4.1.3 With the same conditions as 3.4.1.2, remove the +35 V dc from the +35 V dc in TUNE terminal. The Alarm terminal shall remain grounded. The INHIBIT terminal shall remain open-circuited. The Carrier Insert terminal shall show an open circuit. After a short delay, Power Attenuator Insert terminal shall show an open circuit, and the Tune Complete terminal shall show less than 1 ohm to ground. The time interval between appearance of the open circuit at the Carrier Insert terminal and the open circuit at the Power Attenuator Insert terminal shall be not less than 500 milliseconds (ms) and not more than 3 seconds.

3.4.1.4 With the same conditions existing at the completion of 3.4.1.3, again apply +35 V dc to the +35 V dc in TUNE terminal, +35 V dc shall appear at the INHIBIT terminal, the Alarm terminal shall remain grounded, the Power

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Attenuator Insert, Carrier Insert and Tune Complete terminals shall show an open circuit.

3.4.1.5 With the conditions of 3.4.1.4, remove the +35 V dc from the +35 V dc in TUNE terminal. The INHIBIT terminal shall show an open circuit. Now, remove ground from the OPERATE terminal, +35 V dc shall appear at the INHIBIT and CARRIER INSERT terminals and the Tune Complete terminal shall show an open circuit.

3.4.1.6 With the conditions remaining from 3.4.1.5, again apply a ground to the Operate terminal. The INHIBIT terminal shall show an open circuit. Now apply a ground to the TUNE ACTIVATE terminal, +35 V dc shall appear at the INHIBIT terminal.

3.4.1.7 Remove all inputs. The Alarm terminal shall show an open circuit.

3.5 Design requirements. The Interface Adapter shall:

(a) Provide the necessary control interface between the AN/SRA-57 and 58 antenna coupler groups and either the MX-4845/SRA-34(V) or the AN/SRC-16 NTDS system.
(b) Mount entirely within an AN/SRA-57 or 58 coupler group to maintain the existing external dimensions and cable entrances on the AN/SRA-57 and 58.
(c) Require no circuitry or wiring changes to equipments supplied with the AN/SRC-23(V), AN/SRA-34(V), or AN/SRC-16.
(d) Exercise positive control over transmitter output such that RF switching within the AN/SRA-57 or 58 or under RF power is prevented in normal operation.

3.5.1 Environmental conditions. The equipment shall be capable of operation under only the following conditions as specified under MIL-E-16400 or as modified herein:

(a) Temperature - Class 4
(b) Humidity
(c) Shock - Grade A, Class III Lightweight, Type A Equipment
(d) Inclination
(e) Vibration - Type I Equipment

3.5.1.1 Electromagnetic interference (EMI). Use of the interface adapter shall not compromise the EMI/EMC integrity of the host unit beyond the limits of the latter's applicable EMI/EMC specification requirements.

3.5.2 Electrical design. The Interface Adapter shall be powered by the +35 V dc source derived from the AN/SRA-57 or 58 coupler group via the +35 V via CONTROL RELAY line.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of

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all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities, suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system requirement. The contractor shall provide and maintain a written inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.1.2 Quality plan. A quality plan applicable to the equipments covered by this specification shall be prepared for Government approval (see 6.3).

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection system are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The contractor shall prepare test procedures for Government approval covering first article and quality conformance inspection (see 6.3).

4.1.5 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection

(1) Production inspection (see 4.4.1)
(2) Environmental inspection (see 4.4.2)
(3) Inspection of preparation for delivery (see 4.5)

4.3 First article inspection. Unless otherwise specified (see 6.2) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the examination and testing specified in production inspection, environmental inspection, and table I.

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Table I. First article inspection.

Inspection	Reference paragraph
Design requirements	3.5
Electromagnetic interference	3.5.1.1

4.3.1 First article test report. A first article test report shall be prepared covering the results of the contractors first article (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations approved in accordance with 4.1.4 and shall include the requirements of table II.

Table II. Production inspection.

Inspection	Requirement paragraph
Surface examination	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.3.6,
Operating test	3.4.1 and subparagraphs

4.4.2 Environmental inspection. Environmental inspection shall be as listed in table III and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be approved in accordance with 4.1.4.

Table III. Environmental inspection.

Inspection	Requirement paragraph
Temperature - Class 4	3.5.1(a)
Humidity	3.5.1(b)
Shock	3.5.1(c)
Vibration	3.5.1(e)
Inclination	3.5.1(d)

4.4.2.1 Environmental test samples. Samples for the environmental inspection shall be randomly selected in accordance with the following:

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(a) One sample from the first 25 production units fabricated
(b) One sample from each successive 75 units produced, or fraction—thereof

4.4.2.2 Nonconforming environmental sample units. If a sample unit fails the test specified in 4.4.2, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials and processes. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.5).

4.4.2.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Inspection for preparation for delivery. Inspections shall be conducted to insure conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The AN/SRA-57, 58 series of manually tuned HF (2-30 MHz) 2kW peak envelope power (PEP) transmitter multicouplers are intended for use with such general purpose Navy communications transmitters as the AN/URT-23(V). However, the need for use of the manually tuned multicouplers (AN/SRA-57, 58) with coupler adapter MX-4845( )/SRA-34(V) for existing NTDS, which employ automatically tuned receiver-transmitter radio sets AN/SRC-23A(V) has arisen. Because the AN/SRA-57, 58 multicouplers were not specifically designed for use with the AN/SRC-23A(V) transmitter, these multicouplers lack certain basic control functions for proper interface requirements.

6.1.1 In order to use the AN/SRA-57, 58 series of multicouplers with the automatically tuned Radio Set AN/SRC-23A(V), these multicouplers must properly interface with the controlling Coupler Adapter MX-4845/SRA-34(V), (see Military publication 0967-176-3010).

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6.1.2 Certain functions are lacking with the AN/SRA-57, 58 multicouplers as issued:

(a) Information allowing the SRC-23 to know whether the multicoupler is in a TUNE or OPERATE condition.
(b) A time delay in going from TUNE to OPERATE at the multicoupler so that this information can be fed to the MX-4845 and received back at the SRC-23.
(c) Information allowing the SRC-23 to know when the multicoupler is in a fault (ALARM) condition.

(d) Revision of the power attenuator in the AN/SRA-56, 57, Coupler Group to allow operation at 150 watts. (NTDS requirement).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3
(c) Levels of preservation, packaging, packing and marking required (see 5.1)

6.3 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Quality plan (see 4.1.2)
(b) Test procedures (see 4.1.4)
(c) First article test report (see 4.3.1)

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer, on the adequacy of corrective action.

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6.5.1 Initial shipments. Initial shipments of end items shall be contingent upon successful completion of the environmental inspection.

Preparing activity:
Navy - EC

(Project 5895-N188)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 66 WHICH MAY BE USED.

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